Specialization, risk, and causal reasoning

A short toolkit for the ECON 342 final project

Part 1: Ma and Ramondo's paper and a Canadian decision.

Part 2: causal reasoning from the beginning, then a simple comparison.

This new material will not be tested on Midterm 2.
It supports your AI-enhanced analysis and final project.

A productive cluster can also expose a household to risk

People choose a skill, industry, and place **before** learning which activities will have a good year.

1. A productive local industry attracts workers.
2. A larger cluster can become more productive.
3. A household values the average reward and the risk of a bad outcome.

In the paper's baseline, workers stay in their chosen industry and place after the shock. Wages adjust; households have no saving or financial insurance. Trade opens markets and alternative sources of supply. Its effect on **real income risk** depends on wages and consumer prices together.

Source: Ma and Ramondo, NBER Working Paper 35942, October 2026, pp. 6–15.

A higher average can come with a less attractive gamble

Invented real-consumption units. Each state has probability one-half.

Plan	Good	Bad	Average	Certainty equivalent
Specialized	120	60	90	80
Stable	84	84	84	84
Specialized, assumed 10% gain	132	66	99	88

The **certainty equivalent** is the sure amount this person values as highly as the gamble. It depends on preferences.

$$\underbrace{CE = \frac{1}{0.5/c_{\text{good}} + 0.5/c_{\text{bad}}}}_{\text{one illustrative risk preference}}$$

The stable plan beats the first gamble. An *assumed* gain in both states reverses that choice.

Source: Classroom illustration, not Canadian data or a calibrated result. The 10% gain is imposed on real consumption.

Local external scale and a firm's fixed cost

Our earlier firm example

One firm spreads a design or setup bill over more output.

$$AC = c + F/q$$

A larger run lowers fixed cost per unit.

This paper's mechanism

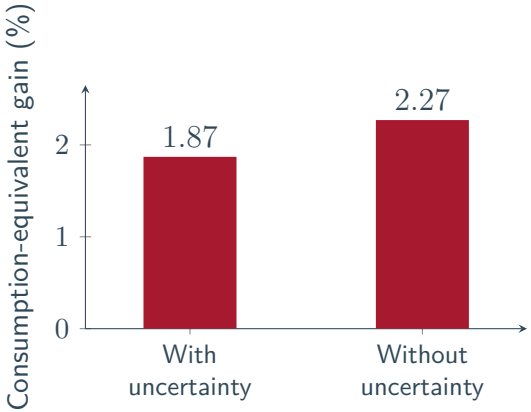
A larger industry *in one place* raises local industry productivity. Shared skills,

suppliers, and knowledge help us picture this external scale effect.

Risk can discourage workers from choosing a cluster in advance. A smaller cluster can then lose some productivity benefits of scale.

Source: Ma and Ramondo, pp. 8–13. The strength of this feedback depends on the model's assumptions and parameters.

Uncertainty reduces the modeled gain from trade

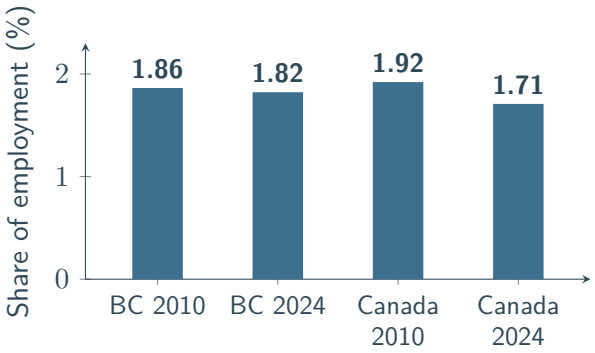


0.40 percentage points

About 18% less than the gain without uncertainty. These are **U.S. model comparisons with autarky**, expressed in welfare units. Every U.S. commuting zone gains in the latest baseline. Ideal national risk sharing largely removes uncertainty's aggregate effect.

Source: Ma and Ramondo, October 2026, Tables 6, 7, and 10. These numbers are neither Canadian estimates nor measured GDP growth.

BC resource employment grew while its share stayed similar



BC's published resource-group employment rose from **41,400 to 53,100 people**. Total employment also grew. The group's share was roughly 1.8–1.9% in both years. This describes the industrial mix. It does not establish a productivity benefit, export advantage, or causal effect.

Source: Statistics Canada 14-10-0023-01, annual employment, ages 15+, total gender. Resource group combines forestry, fishing, mining, quarrying, oil and gas. Denominator: sum of the 16 non-overlapping course groups. Rounded estimates.

The Canadian decision for your project

Choose one province and a clearly defined trade-exposed cluster. Compare it with two provinces and Canada.

- Should the province deepen a specialization, develop a feasible adjacent activity, or preserve the cluster while protecting workers from shocks?
- Exhibit 1: the industrial mix and actual trade exposure.
- Exhibit 2: historical variability, baseline and reweighted mixes, and sensitivity checks.

Employment shares describe concentration. Historical variation describes past outcomes. Add evidence on skills, suppliers, wages, prices, or adjustment to explain the welfare mechanism. **Your conclusion should be conditional:** name the evidence that would change your recommendation.

Source: Project handbook and starter data are linked on the final slide. This toolkit adds no deadline or assessment requirement.

Part 2
Causal reasoning from the beginning

Causal reasoning asks what would change if we did something differently, and what else could explain the pattern.

We start with no prior knowledge and use four numbers to learn one possible method.
For the final project. This new material will not be tested on Midterm 2.

This workshop is separate from the podcast and from the paper's model.

A causal claim says that one thing changes another

A new trade policy begins, and employment in a Canadian industry later falls.

- **Association:** two things move together in the data. The policy began, and employment fell.
- **Causal question:** would changing the policy change employment, compared with what would happen without it?

Causal inference uses data and a justified comparison to learn what effect a change had. A policy decision often needs this evidence.

Source: Hypothetical Canadian example, not a claim about an observed policy.

Other forces can explain the pattern

Suppose world demand for a resource falls.

1. Firms sell less and hire fewer workers.
2. The same decline prompts demands for trade protection.
3. Protection and job losses now appear together.

A confounder is another factor that helps explain both the policy and the outcome. Here it is weaker world demand. To learn what the policy changes, we need to account for such alternative explanations.

Source: Hypothetical mechanism. Other possibilities include technology, commodity prices and the business cycle.

The counterfactual is the missing outcome

Counterfactual: employment in the same group, at the same later time, if the policy had not happened.

We observe	**We want to compare it with**
Employment after the policy, in the world where it happened.	Employment at that same time, in the world without the policy.

We cannot observe both worlds for the same group. Earlier employment is a useful baseline, but demand and other conditions may change over time. A carefully chosen comparison group may help estimate the missing outcome.

Groups, outcomes, and timing

- **Outcome:** the measured result; here, the employment rate, the percentage of the chosen population in work.
- **Treatment:** the policy being studied.
- **Treated group:** the regions or people affected by that policy.
- **Comparison group:** a group not directly affected, used to track shared changes over time.

Choose a **before period** and an **after period**. Keep the outcome definition and population consistent. The groups can start at different levels. We will compare their **changes**.

Differences-in-differences, one step at a time

Invented employment rates. Percentage in work in each group.

Group	Before	After	Change
Treated	70%	64%	$64 - 70 = -6$
Comparison	60%	58%	$58 - 60 = -2$

1. The treated group's rate fell by **6 percentage points**.
2. The comparison group's rate fell by **2 percentage points**.
3. Subtract the changes: $(-6) - (-2) = \boxed{-4}$ percentage points.

Differences-in-differences (DiD) compares the changes in two groups before and after a policy. A percentage point is the difference between two percentages: 70% to 64% is a fall of 6 points.

Source: Four invented numbers. The arithmetic becomes a causal estimate only under a credible comparison and the assumptions explained next.

The assumption behind the four-point difference

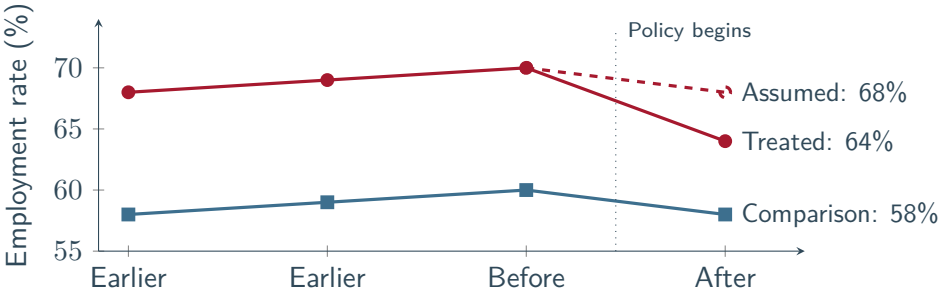
Assume that, **without the policy**, the treated group's employment rate would have changed by the same amount as the comparison group's rate.

$$\underbrace{70\%}_{\text{treated before}} + \underbrace{(58-60)\text{ points}}_{\text{comparison's change}} = \underbrace{68\%}_{\text{assumed without policy}}$$

We observe 64%. Relative to that assumed 68%, the policy's estimated effect is $64-68=-4$ percentage points. The name for the common change *without the policy* is **parallel trends**. It does not require equal employment levels.

Source: The assumed change is the comparison group's contemporaneous fall of 2 points, not an extrapolation of the earlier upward slope.

The dashed path shows the missing comparison



The comparison falls by 2 points. The dashed path gives the treated group the same fall: $70\% \rightarrow 68\%$. The observed endpoint is 4 points lower.

Source: Invented illustration. The dashed segment is an assumption, not an observed outcome. Earlier similar trends can support plausibility but cannot prove this missing path.

What makes the comparison credible?

Ask whether the groups would really have shared the same change without the policy.

- **Other shocks:** did an unrelated factory closure or demand shock affect only one group?
- **Early responses:** did firms act before the chosen policy date? This is anticipation.
- **Effects across groups:** did supply chains or retaliation change the comparison group? These are spillovers.
- **Consistent measurement:** did the population, survey, or definition of employment change differently?

Similar earlier trends help us judge the comparison. They cannot prove what would have happened later without the policy.

Source: Roth et al. (2023). Policy details, earlier data and alternative comparisons all matter.

Claims and evidence in the final project

- **Description:** "Employment fell after the tariff." Check the data, timing and units. This describes what happened.
- **Causation:** "The tariff caused employment to fall." This needs support for the comparison with employment without the tariff.
- **Conditional scenario:** "Holding each industry's observed historical growth rates unchanged, this alternative mix has lower historical variability." State the assumption and what it leaves out.

A model explains a possible channel. A regression summarizes patterns. A before-and-after change measures a change. **Each needs a credible missing-world comparison to establish causation.**

Source: The paper's model simulations and the handbook's fixed-mix calculations are conditional exercises, not DiD estimates.

Evidence options for your final project

If your recommendation relies on **X causing Y**, support that link with:

- a relevant published causal study: explain its comparison, assumptions and fit to your Canadian case; or
- your own credible comparison, if feasible. DiD is one possible tool.

Check the design, measurement, alternative explanations and uncertainty. Narrow the claim when evidence is weak. **A DiD analysis is not required for every project.**

Copyable Codex prompt

For [my Canadian question], define the policy, affected people, measured outcome, and time period in plain English. Which claims are causal? Find a relevant credible causal study or propose a comparison. Explain what would happen without the policy and the assumptions needed. Check other shocks, early responses, and effects on the comparison group. If evidence is weak, narrow the claim. Verify sources; invent no data.

Readings, data, and project resources

- Ma and Ramondo, *Comparative Advantage and Economies of Scale under Uncertainty*, October 2026. Model pp. 6–15; Tables 6, 7, and 10.
- Statistics Canada 14-10-0023-01: annual industry employment.
- Callaway: a two-period DiD introduction; Roth et al. (2023): review of DiD methods and assumptions.
- Canadian specialization project handbook and starter data ZIP.
- Podcast, transcript, chapters, and previous-course review.

This new paper and causal-reasoning workshop will not be tested on Midterm 2. They support your final project. Use causal tools when your claim needs them; no universal requirement to run DiD is added.